Enterprise Infrastructure Planning

NovaTech Startup Solutions

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This plan proposes the initial IT environment for a fictional 20-person software company. It connects employee needs to equipment, software, network access, backup and support responsibilities before purchasing or deployment. The design uses 14 desktops, six laptops, a Linux server and a segmented office network, with quantities and operational responsibilities documented for review.

Part 1 Company Profile

Company name: NovaTech Startup Solutions

Nature of business: A fictional software development startup building web and mobile applications for small businesses.

Vision: To become a trusted software solutions provider that helps small businesses grow through technology.

Mission: To deliver reliable, secure, and affordable software while developing employee skills and protecting customer information.

Office location: 3rd Floor, Meridian Business Center, Brgy. Real, Calamba City, Laguna (fictional).

Office arrangement: One office floor with departmental work areas and a locked equipment area. The CEO and department heads are included in the 20 employees, not additional positions.

Organizational structure

CEO / General Manager → IT Manager, HR Manager, Finance Manager and Sales Manager → staff within each department. Management roles are included in the headcounts below.

Department	Employees	Workstations
Information Technology	5	5 desktops
Human Resources	4	4 desktops
Finance	5	5 desktops
Sales	6	6 laptops
Total	20	14 desktops and 6 laptops

Enterprise Hardware Inventory

The inventory provides one workstation per employee. Shared infrastructure is sized for the single office floor. Asset IDs identify purchasing groups; assign a unique serial-numbered asset record to every physical unit when received. These are proposed specifications, not acquired equipment.

Asset ID	Hardware	Qty	Department	Purpose and planned specification
HW-001	Desktop computer	5	IT	8-core class CPU, 32 GB RAM, 1 TB SSD; development, testing and administration.
HW-002	Desktop computer	4	HR	6-core class CPU, 16 GB RAM, 512 GB SSD; employee records and office work.
HW-003	Desktop computer	5	Finance	6-core class CPU, 16 GB RAM, 512 GB SSD; accounting and financial reports.
HW-004	Laptop	6	Sales	16 GB RAM, 512 GB SSD, Wi-Fi and full-disk encryption; client visits and mobile work.
HW-005	Tower server	1	Shared / IT	8 cores, 32 GB ECC RAM, mirrored boot SSDs and mirrored data disks; files, Git and internal apps.
HW-006	VPN-capable router	1	Shared	WAN routing and provider handoff; avoid duplicate NAT with the firewall.
HW-007	Managed 24-port switch	2	Shared	Gigabit VLAN-capable core SW1 and access SW2; SW1 supplies PoE for two APs.
HW-008	Multifunction printer	2	Shared	One for IT/HR and one for Finance/Sales; network print and scan.
HW-009	UPS	3	Shared	Server/NAS, network edge/core, and access switch/AP path; size from measured load.
HW-010	Wireless access point	2	Shared	Business dual-band Wi-Fi, multiple SSIDs and VLAN tagging; coverage verified on site.
HW-011	NAS storage	1	Shared / IT	Two-bay mirrored storage, at least 4 TB usable initially; restricted local backups.
HW-012	External backup drive	2	IT	Encrypted 4 TB drives rotated weekly; one disconnected and stored off-site.
HW-013	Monitor	16	IT / HR / Finance	14 primary monitors plus two extra displays for IT development and administration.

Selection rationale: fixed-desk departments receive desktops; Sales receives portable laptops. Sixteen monitors cover the 14 desktops and two additional IT displays. The three UPS units protect shared services and network equipment, not every desk. Network equipment in Parts 2 and 4 is the same equipment, not a second purchase. Final UPS capacity and backup-drive size depend on measured load and data growth.

Enterprise Software Inventory

Install only approved, supported releases and record exact build numbers at deployment. Versions below are planning baselines, not a claim that software has been installed. Windows Pro device licensing and Microsoft 365 subscriptions are separate purchases.

Software	Version	License	Purpose
Windows 11 Pro	Supported release	20 OEM/device licenses	Business desktop OS, disk encryption and workstation administration.
Ubuntu Server	24.04 LTS	Open source	Server OS for Samba file shares, Git access, SSH and internal applications. [1,2]
Microsoft Office	Microsoft 365 Apps	20 Business Premium user subscriptions	Word, Excel, PowerPoint and Outlook; business email and collaboration. [3]
VS Code	Approved stable build	Microsoft distribution license; no purchase fee	Code editing, debugging and development extensions for IT.
Git	Approved stable build	Open source (GPLv2)	Version history and collaboration for application code and configuration.
GitHub Desktop	Approved stable build	Open source (MIT)	Graphical Git workflow for IT; private repository access follows company policy.
VirtualBox	Supported 7.x base package	Open-source base package	Isolated development/test VMs on IT PCs. Optional Extension Pack excluded. [4]
Google Chrome	Approved stable build	Proprietary; no purchase fee	Browser-based business services and web application compatibility tests.
Microsoft Defender	Updated signatures / agent	Built-in AV; business management via Premium	Endpoint antivirus with centralized visibility using Defender for Business and Intune. [3]
AnyDesk	Supported business build	Paid commercial license	Authorized remote support by IT; no personal-use license for company support. [5]
7-Zip	Approved stable build	Open source; component licenses apply	Archive and extract files for routine document and backup transfers.

License planning: allocate one Windows Pro license per workstation and one Microsoft 365 subscription per employee. Scope AnyDesk licensing to the IT support team under the selected commercial plan. Review VirtualBox Extension Pack terms separately if its features are later required. Maintain a license register with purchase, assignment and renewal records. [3-5]

Enterprise Network Inventory

Item	Specification	Qty	Purpose
ISP modem / ONT	Provider-managed fiber handoff	1	Business fiber termination; bridge/routed handoff coordinated with ISP.
Router	Gigabit WAN/LAN and VPN capability	1	Provider-facing routing; firewall handles LAN policy, NAT and remote VPN.
Firewall	NGFW with VLAN interfaces, VPN and logging	1	Default-deny inter-VLAN rules, outbound control and inspected remote access.
Managed switch	24-port Gigabit, 802.1Q; PoE on SW1	2	Core SW1 and access SW2; VLAN trunks and wired endpoint connections.
Wireless AP	Dual-band, VLAN-aware business AP	2	Sales and guest coverage; PoE-powered from SW1.
Patch panel	24-port Cat6	1	Terminates 20 planned horizontal office cable runs; four spare positions.
Cat6 cable / patch lead	Correctly rated runs and patch leads	70	20 fixed runs, 20 cabinet leads, 18 endpoint leads, 6 device links and 6 spares.
RJ45 connector	Cat6-compatible plug; match cable type	100	Spare stock for patch-lead fabrication and repairs, not mandatory use on every run.

VLAN and addressing plan

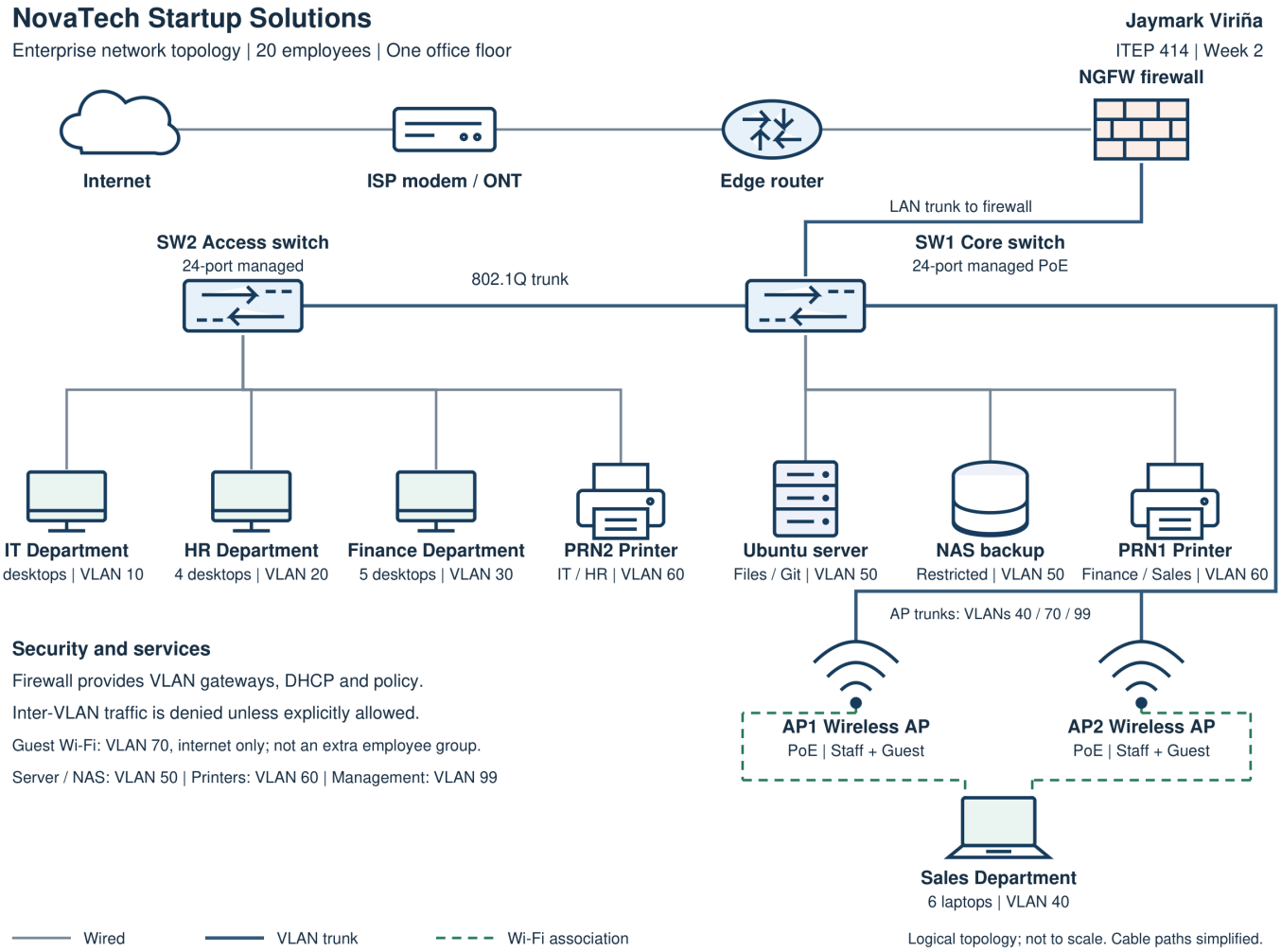
The firewall is the gateway at .1 in each subnet and serves departmental DHCP. Use .100-.199 for user leases; reserve infrastructure addresses. VLAN 99 manages devices and is not a normal user network.

VLAN	Group	Subnet	Intended access / addressing
10	IT	192.168.10.0/24	5 PCs; privileged access only from approved admin devices.
20	HR	192.168.20.0/24	4 PCs; access only to approved HR shares and services.
30	Finance	192.168.30.0/24	5 PCs; restricted accounting and finance shares.
40	Sales	192.168.40.0/24	6 laptops on staff Wi-Fi; approved files and printers.
50	Servers / backup	192.168.50.0/24	Server .10 and NAS .20; backup administration restricted.
60	Printers	192.168.60.0/24	PRN1 .11 and PRN2 .12; print access without broad LAN reach.
70	Guest Wi-Fi	192.168.70.0/24	Internet only with client isolation; blocked from private subnets.
99	Management	192.168.99.0/24	SW1 .11, SW2 .12, AP1 .21, AP2 .22; IT admin access only.

Access policy: permit approved departmental shares and printer services; deny direct department-to-department access. Only approved IT admin devices may manage infrastructure. Guest traffic can reach the internet only. SW1-SW2 and firewall-SW1 are tagged trunks; AP uplinks carry VLANs 40, 70 and 99. Restrict the NAS to backup/admin flows. Terminate fixed Cat6 runs on jacks and the patch panel; use connector types appropriate to the cable.

Enterprise Network Diagram

The topology includes every required device and all four departments. Both switches, both printers and both APs match the inventory. Server, NAS, printers and APs have their own switch connections. Guest Wi-Fi is a separate service; it does not add to the employee headcount.



Connection and capacity notes

SW1 connects to the firewall, SW2, server, NAS, PRN1 and two APs: 7 used ports. SW2 connects 14 desktops, PRN2 and its uplink: 16 used ports. Sales laptops use Wi-Fi rather than adding six required wired endpoints. The two APs provide coverage for the same six laptops; their connections show possible association rather than simultaneous use of both APs.

The edge router handles provider routing and the firewall handles LAN NAT, security and remote VPN. Coordinate a routed handoff or bridge/pass-through arrangement with the ISP to avoid an accidental double-NAT design. Physical cable routes, wireless placement and RF capacity require an office survey before installation.

Editable source: diagrams/NovaTech_Network_Diagram.drawio. Image and print copies are provided as PNG and PDF. The patch panel is passive infrastructure and is omitted from this logical topology. [13]

System Administration Roles

Helpdesk Technician

The helpdesk is the first contact for user incidents. At NovaTech, this role restores everyday productivity and records information needed for escalation. The support tasks below apply entry-level troubleshooting principles to the startup scenario. [6]

Responsibilities: Record and prioritize tickets; install approved workstation software; troubleshoot login, printing and connectivity issues; prepare devices for new staff; verify identity before resets; escalate unresolved incidents with logs and actions already tried.

Skills: Clear communication, patient questioning, Windows troubleshooting, hardware basics, safe remote support, basic network checks and accurate documentation.

Common tools: A ticket queue such as Jira Service Management; licensed AnyDesk; Windows Event Viewer; Intune; ping and ipconfig. Tool choices are proposed for this company.

Certifications: CompTIA A+ is a foundational support certification; the Google IT Support Certificate is another entry-level learning pathway. These are examples for career development, not credentials claimed by the author. [6]

Network Administrator

The network administrator keeps connectivity available and secure. Cisco describes this work as installing, maintaining, monitoring and troubleshooting organizational networks. [7]

Responsibilities: Configure router, firewall, switches and APs; maintain VLANs, DHCP scopes and DNS forwarding; apply access rules; monitor uplinks; save device configurations; troubleshoot wired, wireless and VPN connectivity.

Skills: TCP/IP, subnetting, VLAN tagging, routing, firewall policy, Wi-Fi coverage analysis and systematic fault isolation.

Common tools: Device management interfaces and CLI; Wireshark for packet analysis; an SNMP monitoring system; Cisco Packet Tracer for learning and design exercises.

Certifications: Cisco CCNA for networking fundamentals, IP connectivity, security and automation knowledge. CompTIA Network+ is another possible foundation. [8]

Linux and Cloud Administration

Linux System Administrator

The Linux administrator operates NovaTech's Ubuntu server and protects its data. Ubuntu documentation and Red Hat's RHCSA skill areas support the proposed focus on services, users, storage and security. [1,9]

Responsibilities: Maintain users and groups; apply updates; manage Samba permissions, Git access and internal services; monitor disk space and logs; automate scheduled backups; test restores; maintain recovery documentation.

Skills: Linux command-line work, Bash scripting, file permissions, service management, SSH, storage concepts and backup troubleshooting.

Common tools: OpenSSH, Bash, systemctl, journalctl, apt, rsync and systemd timers. Protect backup credentials separately from ordinary user accounts.

Certifications: Red Hat Certified System Administrator (RHCSA) develops practical Linux administration skills. RHCSA is assessed on Red Hat Enterprise Linux, while the proposed company server runs Ubuntu. [9]

Cloud Administrator

The cloud administrator manages hosted identities, storage, access and cost controls. Microsoft's Azure Administrator role covers compute, networking, storage, identity, security and governance. [10]

Responsibilities: Maintain business cloud accounts; apply least-privilege access and MFA; monitor backup storage and recovery; control subscriptions and spending; review service health; coordinate future migration with the Linux and network administrators.

Skills: Cloud service models, identity and access management, virtual networking, backup retention, resource tagging, cost monitoring and incident coordination.

Common tools: Microsoft 365 admin center and Entra admin center for planned subscriptions; Azure portal and Azure CLI if Azure resources are approved; service monitoring and budget alerts.

Certifications: Microsoft Certified: Azure Administrator Associate is relevant to administering Azure infrastructure. It does not replace product-specific training for Microsoft 365 administration. [10]

How the four professionals work together

When an employee cannot open a shared folder, the Helpdesk Technician records the symptoms, checks the account and tests the connection. If the network path or VLAN rule is responsible, the Network Administrator investigates; if the share or file permissions are responsible, the Linux System Administrator restores access. The Cloud Administrator checks cloud identity, backup availability or a hosted-service outage when those services are involved. All four use the same incident record, agree on ownership, document changes and verify the result with the employee. In a 20-person startup, these responsibilities may be combined among IT staff or supported by a service provider rather than requiring four additional employees.

Infrastructure Recommendations

Internet provider

Request site-specific business-fiber proposals from PLDT Enterprise and Converge Business, starting with a proposed 300 Mbps service. Compare verified coverage, upload speed, uptime commitments, repair response, installation cost and static-IP options. No provider, price, static IP or service-level agreement is assumed confirmed. Schedule large backups outside busy hours and add a separate mobile failover service if downtime would interrupt client work.

Server specifications

Start with an 8-core CPU, 32 GB ECC RAM with an upgrade path, two 480 GB SSDs mirrored for the OS and two 4 TB disks mirrored for data. Use Ubuntu Server 24.04 LTS with updates, least-privilege groups and monitored storage. Its published support lifecycle supports a stable deployment choice. These are planning assumptions for file sharing, Git and small internal services, to be validated with workload measurements. Disk mirroring improves availability but does not replace backups. [1,2]

Backup strategy

Keep production data on the server, a nightly versioned backup on the NAS and a weekly encrypted copy on rotating USB drives, with one drive disconnected off-site. This provides multiple copies across network storage and removable media. Add an encrypted cloud copy of critical data for shorter off-site recovery gaps. Proposed targets are a 24-hour recovery point and an 8-hour recovery time, subject to a restore trial; off-site recovery may lose up to one week without cloud replication. Retain 30 daily and 12 monthly versions where capacity permits, review failures daily and test a sample restore monthly and full recovery quarterly. [11]

Security and Growth

Security recommendations

Enforce VLAN boundaries at the firewall and use the permissions listed in Part 4. Permit SSH only from approved IT administration devices; use keys and disable direct root login. Require VPN with MFA for remote administrative access; do not publish SSH or file shares directly to the internet. Apply device encryption, patching, least privilege, a locked equipment area and configuration backups. Inspect encrypted traffic only under an approved company policy.

Antivirus

Keep Microsoft Defender Antivirus active on every Windows endpoint. Budget Microsoft 365 Business Premium for 20 users to provide the proposed Office apps, Intune management and Defender for Business capabilities. Enroll devices and test alerts; built-in antivirus alone does not establish centralized management. Review detections, isolate affected devices and document recovery actions. [3]

Password policy

Adopt unique passphrases of at least 15 characters, MFA for email, cloud administration and VPN, and a company password manager. Block known compromised passwords; avoid routine forced changes and arbitrary character-composition rules. Change credentials after suspected compromise, protect recovery methods and separate ordinary from administrator accounts. This company policy is informed by NIST guidance and must be implemented within each product's supported controls. [12]

Expansion plan

The initial core switch uses 7 ports and the access switch uses 16, leaving 25 of 48 ports unused. Each of 14 desktops has one cable run; six laptops use Wi-Fi. The 20 horizontal runs comprise 14 desktops, two printers, two APs and two spare outlets. Add staff only after checking PoE capacity, Wi-Fi load, patch-panel capacity and server utilization. Add a patch panel, access switch or AP when measurements justify it. Reassess redundancy and cloud hosting as availability needs grow.

Personal Reflection

Before this project, I thought being a System Administrator was mostly about fixing computers when something goes wrong. Planning the infrastructure for NovaTech helped me understand that the work begins much earlier. A system administrator needs to identify what employees do, choose suitable equipment, organize access, and prepare ways to recover important information. A useful plan explains why each decision supports the business instead of simply listing products.

Preparing the inventories helped me connect equipment choices to actual users. The IT, HR, and Finance departments need desktop computers because most of their work takes place at fixed desks. Sales employees need laptops because their work can involve client visits. I also learned that shared equipment, such as printers, switches, storage, and UPS units, needs to be counted and justified. The quantities in the report should agree with the diagram so that another person can understand what must be purchased.

The most challenging part was organizing the network diagram. Choosing labels was easier than deciding how devices should connect and how departments should be separated. I needed to distinguish a physical connection from a security boundary. A switch connects devices, while VLANs and firewall rules control which groups can communicate. Clear connections and consistent labels make the design easier to explain and reduce confusion during deployment.

Planning is important because mistakes become more expensive after equipment is purchased and installed. Missing backup storage, insufficient network ports, or unclear software licenses can delay work and create unexpected costs. I also realized that having a backup is only useful when it can be restored. A recovery plan should identify who checks backup failures and how often restoration is tested.

This project will help me become a better System Administrator by encouraging me to document decisions, check assumptions, and consider maintenance before deployment. Researching the four IT roles also showed me why communication matters. Helpdesk, network, Linux, and cloud responsibilities are connected even when one person performs several of them. I still need more practical experience configuring and testing these systems, but this activity gives me a clearer process to follow when planning an organization's IT environment.

Reflection length: approximately 353 words.

References

Numbered references correspond to citations in the report. Company-specific specifications, access rules and tool selections are proposed design decisions. Source access date: 25 October 2026.

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